InferenceControl · C4

Spend work deliberately.
Keep the budget spent after a crash.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A limited exam-attempt card: restarting the application does not give back an attempt already reserved.

Why it matters: A bounded native action needs durable limits and honest completion status. Restart must not turn one authorized step into unbounded evaluation.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

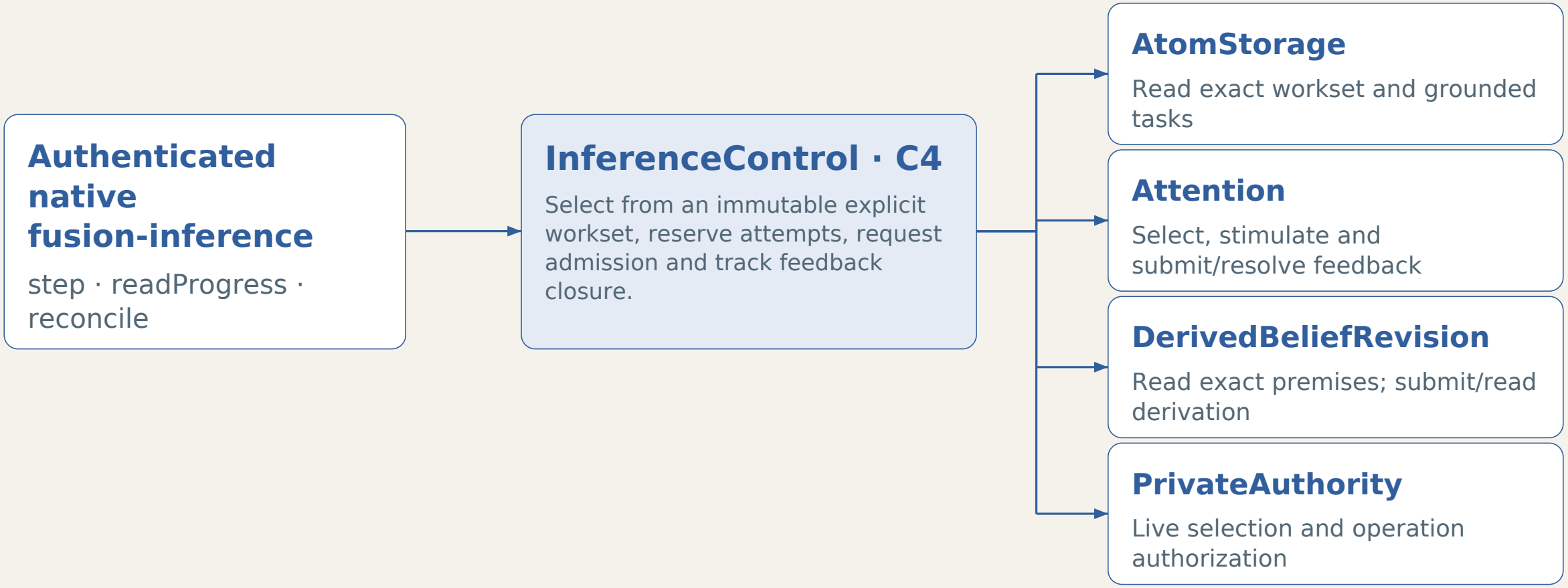
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Select from an immutable explicit workset, reserve attempts, request admission and track feedback closure.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Execution journal

Selections, candidate progress, physical-attempt reservations and immutable result

Closure journal

Bounded feedback reconciliation progress, separate from execution counters

OUTSIDE THIS RESPONSIBILITY

Beliefs and STI

Admission belongs to C2; causal credit belongs to C3

General search/scheduling

No agenda expansion, continuous worker or automatic retry scheduler

bounded-grounded-inference/1; fresh M3.1 local 2.1.0 closure extension

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
step	Run one immutable explicit workset	Budget, deadline and selections bind the intent
readProgress	Expose execution and closure independently	Terminal execution may have pending closure
reconcile	Resolve already-recorded feedback intents	No new inference, retiming, refunds or deadline reset

Finite worksets and journals; ≤16 resolutions/reconcile call; absolute deadlines never renewed.

A worked example through the boundary

ACTUAL native Hive inference · generated workset; real C2/PLN/DeTTa

01 · STEP

Freeze selection

Read eligible tasks and exact premise snapshots.

02 · STEP

Reserve

Persist one physical attempt before evaluation.

03 · STEP

Request C2

C2 invokes guarded PLN and decides admission.

04 · STEP

Close separately

Save feedback outcome; later reconcile if needed.

What the example establishes

Bounded work can finish honestly without finding a conclusion.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Durable budgets

Candidate/attempt/generation limits and absolute deadline survive restart.

I2

Physical reservation

A crash may consume a slot without a conclusion; no refund.

I3

Frozen inputs

No silent latest-premise substitution or workset expansion.

I4

Honest progress

Later closure never rewrites the original execution result.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Crash after reservation	Slot remains spent, even if native evaluation never completes.	Conservative bounded-work guarantee
Admission saved; reply lost	Read the recorded C2 decision and continue original intent.	Duplicate-safe support admission
Feedback application reply lost	Resolve at C3; preserve original result and budget.	No new native formula calls

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

2 / 1

Candidates / attempt in comparison

One admission and one feedback credit

0

New native calls during reconciliation

Recorded M3.1 observation

1–16

Tasks in explicit workset

Contract bound, not search depth

The historical four-case comparison records one full step per configuration, not a latency distribution. No sustained scheduling, fairness or general inference benchmark exists.

PROPOSED BENCHMARK · NOT RUN

Proposed: fixed worksets and native runtime; vary budgets, refused candidates and crash points. Report completion/abandonment, physical evaluations and p50/p95 step/closure latency.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Persistent controller, actual authenticated Hive action and native guarded PLN through C2.
FIXTURE / LOCAL TEST POLICY	Generated tasks/premises, initial priority preference and injected kill positions.
BLOCKED / UNSUPPORTED	No required supported gate blocked. Eligible/unavailable feedback can remain pending by design.
UNRUN	General inference search, scheduler fairness, LTI, distributed recovery and old-state migration.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure budget utility	Fixed tasks, declared objective and failure schedule	Useful completions per physical evaluation
Improve closure observability	Read-only authorized receipt tooling	Separate execution, evidence and feedback progress
Specify orchestration separately	Explicit ownership and retry policy before automation	No scheduler hidden inside reconciliation

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Bounded work can finish honestly without finding a conclusion.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m3_1.py <env> --output <fresh>
```

Actual Hive cases require pinned native providers and private operator configuration. Fresh closure journals preserve explicit compatibility; no live migration.

`fusion/inference.py`
`fusion/deadlines.py`
`fusion/cognition_contracts.py`
`integration/hive/inference_program/program.py`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.